

Annual control work, and preparation for the entrance paper

Two years of calculus in one paper, then what the university examination actually asks.

LESSON 101–102

2 × 40 MIN

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ANNUAL REVIEW

LESSON CLOCK

3

50'

12'

10'

5'

Instructions	3 min
The paper	50 min
Answers	12 min
Diagnosis	10 min
The entrance paper	5 min

LEARNING OBJECTIVES

- Apply differentiation, integration, probability and complex numbers in one paper.
- Choose the shortest route to each answer under time pressure.
- Name each lost mark and rewrite the solution in full.
- Recognise the standard question types of the entrance examination.

TEXTBOOK

National	pp. 461–464
Cambridge	Pure Mathematics 2 & 3, pp. 277–280
Workbook	Annual paper A

01

Explanation

The paper — 50 marks, 50 minutes

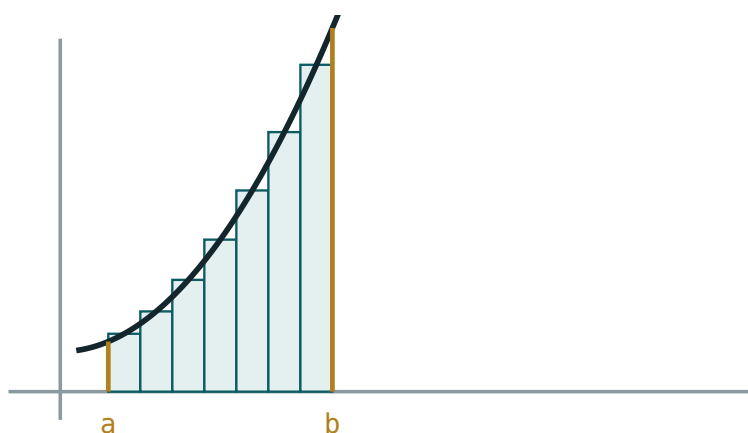
Q	TASK	MARKS	FROM
1	Differentiate $y = x^3 \ln x$ and $y = \frac{\sin x}{x^2}$	6	Q I
2	Find the stationary points of $y = x^3 - 6x^2 + 9x + 2$ and classify them	7	Q I
3	Evaluate $\int_1^4 (2x + \frac{1}{\sqrt{x}}) dx$ and find the area between $y = x^2$ and $y = 2x$	8	Q II–III
4	A box of 4 red and 6 blue: two drawn without replacement — the tree and three probabilities	7	Q IV
5	$X \sim B(10, 0.3)$: μ , σ , $P(X = 2)$, $P(X \geq 1)$	7	Q IV
6	Simplify $\frac{2 + 3i}{1 - i}$, solve $z^2 + 2z + 5 = 0$ and describe $ z - 2i = 3$	8	Q IV
7	A sample decays with half-life 6 days from 400 units: form and solve the equation, then find $N(15)$	7	Q IV

HOW TO SPEND 50 MINUTES

Questions 1, 5 and 6 are mechanical — about 18 minutes for 21 marks. Do them first. Questions 2, 3 and 7 reward care rather than speed. Leave whichever of them you find hardest to last, and write the setting-up even if you cannot finish: method marks are real marks.

The two years in one page

BLOCK	THE ONE IDEA	THE ONE FORMULA
the derivative	the gradient of the tangent	$\frac{dy}{dx} = \lim \frac{\Delta y}{\Delta x}$
applications	$f' = 0$ locates the turning points	f'' decides the kind
the integral	the reverse of the derivative	$\int f' = f + c$
the definite integral	signed area	$\int_a^b f = F(b) - F(a)$
probability	count, or measure, then divide	$P = \frac{m}{n}$
distributions	the shape of many repetitions	$\mu = np$
complex numbers	$i^2 = -1$	$z \bar{z} = z ^2$
differential equations	a statement about a rate	<i>separate and integrate</i>



more, thinner rectangles → the definite integral

*The picture that connects the two halves of the course
— the integral as area.*

The entrance paper

The university entrance examination in mathematics rewards accuracy and speed on standard types rather than ingenuity. The types repeat.

TYPE	WHAT IT ASKS	WHERE IT CAME FROM
algebraic	simplify, solve, factorise	Grades 8–10
function	domain, range, graph, inverse	Grade 10 Q I
trigonometric	equations and identities	Grade 10 Q IV
exponential and log	equations and inequalities	Grade 10 Q III
calculus	tangent, extremum, area	Grade 11 Q I–III
combinatorial	counting and probability	Grade 11 Q III–IV
geometric	a solid, or a coordinate figure	Grade 10–11 geometry

FOUR RULES THAT RAISE A SCORE WITHOUT NEW MATHEMATICS

- **Read to the end** before starting; the last clause often changes the method.
- **Write the setting-up** even when stuck — a , d , n , p , the sketch, the formula.
Method marks do not require an answer.
- **Check the domain**: a root, a logarithm, a denominator, $|t| \leq 1$ — every rejected root is a mark.
- **Leave nothing blank**. A named formula and one substituted line is worth more than white space, and costs twenty seconds.

What to practise over the summer. Twenty past papers, timed, one a day, marked the same evening against the scheme — and a written list of every type that cost a mark twice. That list, not the papers, is what improves the score.

02 Worked examples

EXAMPLE 1 Model answer, Q2: the stationary points of $y = x^3 - 6x^2 + 9x + 2$.

$$① \quad y' = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$$

$$② \quad x = 1 \text{ and } x = 3.$$

$$③ \quad y'' = 6x - 12: y''(1) = -6 < 0, y''(3) = 6 > 0.$$

$$④ \quad \text{Maximum } (1, 6), \text{ minimum } (3, 2).$$

ANSWER

Maximum $(1, 6)$, minimum $(3, 2)$

EXAMPLE 2 Model answer, Q3: the area between $y = x^2$ and $y = 2x$.

$$① \quad x^2 = 2x \Rightarrow x = 0, 2$$

$$② \quad \int_0^2 (2x - x^2) dx$$

Upper minus lower.

$$③ \quad = \left[x^2 - \frac{x^3}{3} \right]_0^2$$

$$④ \quad = 4 - \frac{8}{3} = \frac{4}{3}$$

ANSWER

$$\frac{4}{3}$$

EXAMPLE 3 Model answer, Q6: simplify $\frac{2+3i}{1-i}$ and solve $z^2 + 2z + 5 = 0$.

$$(1) \quad \frac{(2+3i)(1+i)}{2} = \frac{2+2i+3i-3}{2}$$

$$(2) \quad = \frac{-1+5i}{2}$$

$$(3) \quad D = 4 - 20 = -16 \Rightarrow \sqrt{D} = 4i$$

$$(4) \quad z = -1 \pm 2i$$

ANSWER

$$\frac{-1+5i}{2} \text{ and } z = -1 \pm 2i$$

EXAMPLE 4 Model answer, Q7: half-life 6 days from 400 units — find $N(15)$.

$$(1) \quad \frac{dN}{dt} = -kN \Rightarrow N = 400e^{-kt}$$

$$(2) \quad 200 = 400e^{-6k} \Rightarrow k = \frac{\ln 2}{6} \approx 0.1155$$

$$(3) \quad N(15) = 400e^{-1.733}$$

$$(4) \quad \approx 70.7$$

ANSWER

$$\approx 71 \text{ units}$$

03 Interactive model

Hand out the mark allocation before the paper starts; ten seconds of planning is worth five marks, every time.

The two years in twelve questions

d $\frac{d}{dx} (x^3 \ln x)$ is:

$$3x^2 \ln x$$

$$3x^2 \ln x + x^2$$

$$x^2$$

$$3x^2 + \frac{1}{x}$$

A maximum has:

$$f'' > 0$$

$$f'' < 0$$

$$f' > 0$$

$$f'' = 0$$

$\int x^2 dx$ is:

$$2x + c$$

$$\frac{x^3}{3} + c$$

$$x^3 + c$$

$$\frac{x^3}{2} + c$$

$\int_0^2 2x \, dx$ is:

2

4

8

0

Area between curves is:

upper minus lower

lower minus upper

their sum

their product

$P(A \cap B)$ in general:

$P(A)P(B)$

$P(A)P(B | A)$

$P(A) + P(B)$

0

The mean of $B(10, 0.3)$:

3

7

2.1

10

Within two σ lies:

68\%

95\%

99.7\%

50\%

i^2 is:

1

-1

i

0

$|z - a| = r$ is:

a line

a circle

a half-line

a point

The solution of $\frac{dy}{dx} = ky$:

$y = kx + c$

$y = Ae^{kx}$

$y = \ln kx$

$y = kx^2$

Equilibrium is where:

$$y = 0$$

$$\frac{dy}{dx} = 0$$

$$x = 0$$

$$k = 0$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Annual control work	Yillik nazorat ishi	Годовая контрольная работа
Derivative	Hosila	Производная
Integral	Integral	Интеграл
Stationary point	Statsionar nuqta	Стационарная точка
Definite integral	Aniq integral	Определённый интеграл
Probability	Ehtimollik	Вероятность
Complex number	Kompleks son	Комплексное число
Entrance examination	Kirish imtihoni	Вступительный экзамен
Time management	Vaqtini taqsimlash	Распределение времени
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which questions should be done first?

in order

the mechanical ones — 1, 5 and 6

the hardest

the longest

A blank answer earns:

nothing

a method mark for the formula

half marks

a warning

The entrance paper rewards:

ingenuity

accuracy and speed on standard types

long proofs

guessing

The best summer practice is:

reading

timed past papers, marked the same evening

new topics

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{d}{dx}(x^3 \ln x)$

ANSWER $3x^2 \ln x + x^2$

2. $\frac{d}{dx}\left(\frac{\sin x}{x^2}\right)$

ANSWER $\frac{x \cos x - 2 \sin x}{x^3}$

3. $\int_0^2 2x \, dx$

ANSWER 4

4. μ of $B(10, 0.3)$

ANSWER 3

5. σ of $B(10, 0.3)$

ANSWER ≈ 1.45

6. Solve $z^2 + 2z + 5 = 0$

ANSWER $-1 \pm 2i$

7. Describe $|z - 2i| = 3$

ANSWER Circle, centre $2i$, radius 3

Medium · the standard question

7 PROBLEMS

1. Stationary points of $y = x^3 - 6x^2 + 9x + 2$

ANSWER $(1, 6)$ max, $(3, 2)$ min

2. $\int_1^4 (2x + \frac{1}{\sqrt{x}}) dx$

ANSWER $[x^2 + 2\sqrt{x}]_1^4 = 20 - 3 = 17$

3. Area between $y = x^2$ and $y = 2x$

ANSWER $\frac{4}{3}$

4. Box of 4 red, 6 blue, two drawn without replacement: $P(RR)$

ANSWER $\frac{2}{15}$

5. Same: $P(\text{one of each})$

ANSWER $\frac{8}{15}$

6. $P(X = 2)$ for $B(10, 0.3)$

ANSWER ≈ 0.233

7. Simplify $\frac{2 + 3i}{1 - i}$

ANSWER $\frac{-1 + 5i}{2}$

Hard · stretch and reason

7 PROBLEMS

1. $P(X \geq 1)$ for $B(10, 0.3)$

ANSWER ≈ 0.972

2. Half-life 6 days from 400: $N(15)$

ANSWER ≈ 71

3. Greatest and least $|z|$ on $|z - 2i| = 3$

ANSWER 5 and 0 — the origin is inside

4. Volume when $y = x^2$ from 0 to 2 rotates about Ox

ANSWER $\frac{32\pi}{5}$

5. A box of 4 red and 6 blue: $P(\text{second red})$

ANSWER $\frac{2}{5}$

6. Show that $y = x^3 - 6x^2 + 9x + 2$ has an inflection at $x = 2$

ANSWER $y'' = 6x - 12 = 0$

7. Solve $\frac{dy}{dx} = \frac{y}{x^2}$ with $y(1) = e$

ANSWER $y = e^2 e^{-1/x}$

07 Homework

Homework — 4 tasks

This is the last algebra lesson of the school course. The summer list is the part that matters.

1. Rewrite in full every annual-paper question that lost a mark, naming the slip in the margin.
2. Write the two years in one page: the eight blocks with one formula and one sketch each.
3. Work one complete past entrance paper under timed conditions and mark it against the scheme the same evening.

4. Write the list of question types that cost you a mark more than once, and one sentence saying how you will practise each.